Multivariable calculus

1. Vectors

1.1. $\mathbb{R}^n$

Definition 1.1.1 (Two-dimensional real-coordinate space ($\mathbb{R}^2$))

$$\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}$$

Definition 1.1.2 (Three-dimensional real-coordinate space ($\mathbb{R}^3$))

$$\mathbb{R}^3 = \{(x, y, z) : x, y, z \in \mathbb{R}\}$$

x, y, z should be presented such that the coordinate system is right-handed ($\hat{k} = \hat{\imath} \times \hat{\jmath}$ should have direction according to the right-hand rule).

1.2. Vector space

Definition 1.2.1 (Vector space)

A vector space over a field F is a set V with the following operations:

- **Vector addition**: $u \in V, v \in V \mapsto (u + v) \in V$, which satisfies the properties of addition
- **Scalar multiplication**: $\alpha \in F, v \in V \mapsto \alpha v \in V$, which satisfies the properties of scalar multiplication

Definition 1.2.2 (Properties of addition)

For all $u, v, w \in V$:

- (A1) Commutativity: $u + v = v + u$
- (A2) Associativity: $(u + v) + w = u + (v + w)$
- (A3) Identity: $\exists 0 \in \mathbb{R}$ s.t. $0 + u = u$
- (A4) Additive inverse: For $u \in F, \exists -u \in F$ s.t. $u + (-u) = 0$

Definition 1.2.3 (Properties of scalar multiplication)

For all $\alpha, \beta \in F, v, w \in V$:

- (S1) Associativity: $(\alpha\beta)v = \alpha(\beta v)$
- (S2) Distributivity over scalar addition: $(\alpha + \beta)v = \alpha v + \beta v$
- (S3) Distributivity over vector addition: $\alpha(v + w) = \alpha v + \alpha w$
- (S4) Multiplicative identity: $1v = v$

Theorem 1.2.4

$\mathbb{R}^n$ is a vector space over the field $\mathbb{R}^n$, where we define

$$\begin{bmatrix} v_1 \\ v_2 \\ \vdots \end{bmatrix} + \begin{bmatrix} w_1 \\ w_2 \\ \vdots \end{bmatrix} = (v_1 + w_1, v_2 + w_2, \vdots)$$

$$\alpha \begin{bmatrix} v_1 \\ v_2 \\ \vdots \end{bmatrix} = \begin{bmatrix} \alpha v_1 \\ \alpha v_2 \\ \vdots \end{bmatrix}$$

Definition 1.2.5 (Standard basis vectors)

The standard basis vectors of a space are the unit vectors that go along the axes of the space. All vectors in that space can be expressed as sums of scalar multiples of the standard basis vectors of that space.

The standard basis vectors of $\mathbb{R}^2$ are $\hat{\imath}$ and $\hat{\jmath}$, also called $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$.

The standard basis vectors of $\mathbb{R}^3$ are $\hat{\imath}$, $\hat{\jmath}$, and $\hat{k}$, also called $\boldsymbol{e}_1$, $\boldsymbol{e}_2$, and $\boldsymbol{e}_3$.

Definition 1.2.6 (Vector length)

The ***length*** or ***magnitude*** of a vector $\begin{bmatrix} v_1 \\ v_2 \\ \vdots \end{bmatrix}$ in $\mathbb{R}^n$ is

$$\left| \begin{bmatrix} v_1 \\ v_2 \\ \vdots \end{bmatrix} \right| = \left\| \begin{bmatrix} v_1 \\ v_2 \\ \vdots \end{bmatrix} \right\| := \sqrt{v_1^2 + v_2^2 + \cdots}$$

Definition 1.2.7 (Displacement vector)

The vector from the end of $\vec{A}$ to the end of $\vec{B}$ when their starts are in the same location.

$$\overrightarrow{AB} = \vec{B} - \vec{A}$$

1.3. Dot product

Definition 1.3.1 (Dot product)

Where $\vec{a}, \vec{b} \in \mathbb{R}^n$ and θ is the angle between $\vec{a}$ and $\vec{b}$:

$$\vec{a} \cdot \vec{b} = \sum a_i b_i = |\vec{a}| \left|\vec{b}\right| \cos\theta$$

Also called ***scalar product*** or ***inner product***.

Corollary 1.3.1.1

Where $\vec{a}, \vec{b} \in \mathbb{R}^n$ and θ is the angle between $\vec{a}$ and $\vec{b}$:

$$\cos(\theta) = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| \left|\vec{b}\right|}$$

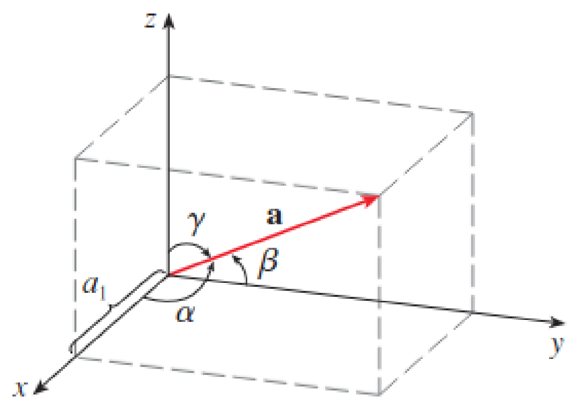
Theorem 1.3.2 (Properties of dot product)

For $\vec{a}, \vec{b}, \vec{c} \in \mathbb{R}^n$ and $k \in \mathbb{R}$:

- $\vec{a} \cdot \vec{a} = |\vec{a}|^2$
- $\vec{a} \cdot \vec{a} = 0$ iff $\vec{a} = 0$
- Commutativity: $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$
- Distributivity: $\vec{a} \cdot \left(\vec{b} + \vec{c}\right) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$
- Distributivity: $(k\vec{a}) \cdot b = k\left(\vec{a} \cdot \vec{b}\right) = a \cdot \left(k\vec{b}\right)$
- $\vec{a} \cdot \vec{b} = 0$ iff $\vec{a} \perp \vec{b}$, $\vec{a} = 0$, or $\vec{b} = 0$.

Definition 1.3.3 (Direction angle)

The ***direction angles*** of a nonzero vector $\vec{a}$ are the angles $\alpha, \beta, \gamma \in [0, \pi]$ that $\vec{a}$ makes with the positive x, y, z axes respectively.



Theorem 1.3.4

The direction cosines of $\vec{a}$ are the components of the unit vector in the direction of $\vec{a}$:

$$\frac{1}{\vec{a}}\vec{a} = \begin{bmatrix} \cos\alpha \\ \cos\beta \\ \cos\gamma \end{bmatrix}.$$

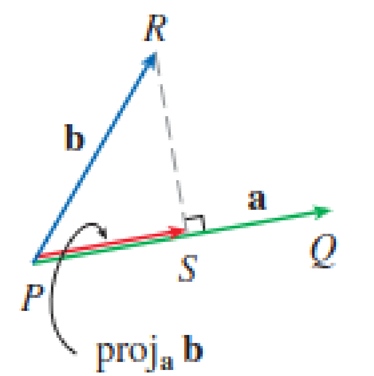
Definition 1.3.5 (Projection)

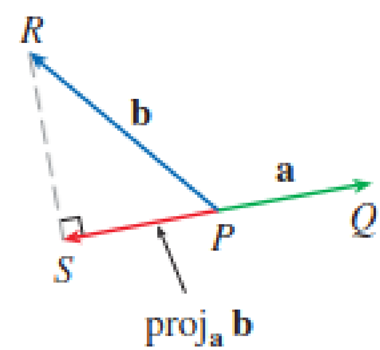
The ***scalar projection of $\vec{b}$ onto $\vec{a}$*** or ***component of $\vec{b}$ along $\vec{a}$***, is defined as (where $\theta \in [0, \pi]$ is the angle between $\vec{a}$ and $\vec{b}$):

$$\text{comp}_{\vec{a}}\vec{b} := \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|} = \left|\vec{b}\right| \cos(\theta)$$

The ***vector projection of $\vec{b}$ onto $\vec{a}$***, is defined as:

$$\text{proj}_{\vec{a}}\vec{b} := \text{comp}_{\vec{a}}\vec{b}\frac{\vec{a}}{|\vec{a}|} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|^2}\vec{a}$$





1.4. Cross product

See Stewart 12.4, NMD Lecture 4

Definition 1.4.1 (Cross product)

For $\vec{a}, \vec{b} \in \mathbb{R}^3$, the unique vector $a \times b$ satisfying

- $|a \times b|$ is the area of the parallelogram spanned by a and b
- $a \times b = 0$ iff $a \parallel b$, $a = 0$, or $b = 0$.
- $a \times b$ is orthogonal to a and b.
- $(a, b, a \times b)$ is right-handed (if the coordinate system is right-handed)

Theorem 1.4.2 (Properties of cross product)

For $a, b, c \in \mathbb{R}^3$ and $k \in \mathbb{R}$:

- $a \times b = (-b) \times a$
- $a \times (b + c) = a \times b + a \times c$
- $(a + b) \times c = a \times c + b \times c$
- $k(a \times b) = (ka) \times b = a \times (kb)$
- $a \cdot (b \times c) = (a \times b) \cdot c$
- $a \times (b \times c) = (a \cdot c)b - (a \cdot b)c$

Theorem 1.4.3 (Calculation of cross product)

Where $a, b \in \mathbb{R}^3$ and θ is the angle between a and b:

$$a \times b = \begin{bmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{bmatrix} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$= \hat{\imath}\begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \hat{\jmath}\begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \hat{k}\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}$$

$$|a \times b| = |a|\ |b| \sin\theta$$

1.4.1. Parallelograms and parallelepipeds

Theorem 1.4.4 (Parallelogram spanned by vectors in $\mathbb{R}^2$)

The area of the parallelogram spanned by the vectors $\vec{v}, \vec{w} \in \mathbb{R}^2$ is given by

$$\begin{vmatrix} v_1 & v_2 \\ w_1 & w_2 \end{vmatrix} = v_1w_2 - v_2w_1.$$

The sign of the result is positive if $\vec{w}$ is counterclockwise from $\vec{v}$ and negative if clockwise.

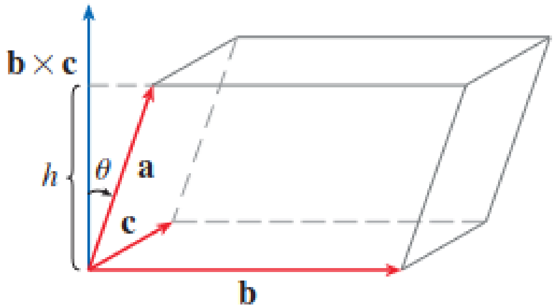
Definition 1.4.5 (Triple product)

The ***triple product*** of $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^3$ is defined as

$$u \cdot (v \times w) = (u \times v) \cdot w = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$$

$$= w_1\begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} - w_2\begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} + w_3\begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}$$

Theorem 1.4.6 (Volume of parallelepiped)



Consider the parallelepiped determined by $\vec{a}, \vec{b}, \vec{c} \in \mathbb{R}^3$, shown above. The area of its base is $|b \times c|$. Let θ be the angle between a and $b \times c$; then, its height is $|a||\cos\theta|$. Thus, the volume of the parallelepiped is

$$V = Ah = |b \times c||a||\cos\theta| = |a \times (b \times c)|.$$

2. Lines and planes

2.1. Lines

The most useful notation for a line is in parametric form:

Definition 2.1.1 (Parametric form of a line)

Where $\vec{r}_0 \in \mathbb{R}^3$ is a point on the line, and $t \in \mathbb{R}^3$:

$$\vec{r}(t) = \vec{r}_0 + \vec{v}t$$

$\vec{v}$ is called the ***direction vector***. Then, every value of t gives a point $\vec{r}(t)$ on the line.

Theorem 2.1.2

Two lines are parallel iff their direction vectors are scalar multiples of each other.

Definition 2.1.3 (Skew lines)

Two lines are skew iff they do not intersect but are not parallel, i.e. they lie in different parallel planes.

Procedure 2.1.4 (Finding the intersection of two lines)

Given lines with parametric equations

$$r_1(t) = a_1 + v_1 t \quad r_2(t) = a_2 + v_2 t$$

solve the system of equations

$$a_1 + v_1 t_1 = a_2 + v_2 t_2$$

for t_1 or t_2, then plug it in to the appropriate equation.

(Break up the equation into its x, y, and z components, or whichever is appropriate for your coordinate space.)

2.2. Planes

See Colley [1.5]

Definition 2.2.1 (Plane)

A plane Π is determined uniquely by a point P in the plane and a normal vector n.

A plane is the set of points A in space such that $\overrightarrow{AP}$ is perpendicular to n.

Theorem 2.2.2 (Scalar equation for a plane in $\mathbb{R}^3$)

If $n \in \mathbb{R}^3$ is the **normal vector** to the plane (vector perpendicular to the plane), and $P \in \mathbb{R}^3$ is a point on the plane:

$$n_x(x - P_x) + n_y(x - P_y) + n_z(z - P_z) = 0$$

or equivalently:

$$n_x x + n_y y + n_z z = n_x P_x + n_y P_y + n_z P_z$$

Procedure 2.2.3 (Equation of plane containing three points)

If $A, B, C \in \mathbb{R}^3$ are points on our plane, then we can find the normal vector by performing $n = \overrightarrow{AB} \times \overrightarrow{AC} = (B - A) \times (C - A)$ (since $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are vectors on the plane).

Theorem 2.2.4 (Parametric equation for a plane in $\mathbb{R}^3$)

If $a, b \in \mathbb{R}^3$ are nonparallel nonzero vectors on the plane, and $P \in \mathbb{R}^3$ is a point on the plane, then the parametric equation for the plane is:

$$x(s, t) = P + sa + tb$$

2.3. Distance

See Colley [1.5]

Procedure 2.3.1 (Distance between point and line)

Let P be the point, and $A + Lt$ be the line. Then the distance is

$$|\overrightarrow{AP} - \operatorname{proj}_L \overrightarrow{AP}| = \vec{P}$$

Procedure 2.3.2 (Distance between parallel planes)

Let Π_1 and Π_2 be the two planes.

If n is normal to both planes, and $P_1 \in \Pi_1$ and $P_2 \in \Pi_2$, then the answer is

$$|\operatorname{proj}_n \overrightarrow{P_1 P_2}|$$